

Numerical Analysis Exercise 5

1. Use Gaussian elimination with partial pivoting to solve the following linear system.(手算)

We can get that by dividing matrices:

```
1  clc;
2  clear all;
3  close all;
4
5  A = [1,1,-1;1,2,4;2,-1,2];
6  %A = input("Please input the coefficient matrix:")
7  b = [1,2,3]';
8  %b = input("Please input the constant matrix:")
9
10 x=A\b
```

```
1  x = 3x1
2      1.2632
3      -0.0526
4      0.2105
5
```

2. Solve the following linear system by using Chasing method(refer to "追赶法.PPT")

$$\begin{cases} x_1 + x_2 & = 5 \\ 2x_1 - x_2 + 5x_3 & = -9 \\ 3x_2 - 4x_3 + 2x_4 & = 19 \\ 2x_3 + 6x_4 & = 2 \end{cases}$$

```
1  %Chasing method
2  clc;
3  clear all;
4  close all;
5  A=[1,2,0,0;2,3,-1,0;0,4,2,-1;0,0,2,-4];
6  b=[7,9,10,12];
7  b=b';
8  %A=input("Please input the A:")
9  %b=input("Please input the b:")
10
11 ab=[A,b];%a为系数矩阵, b为右端项
12 [R,C]=size(ab);
13 for i=1:R-1 %控制消去首行元素的次数 二阶方程组只需要消去一次,三阶消去两次
14     %i=2时, 系数矩阵第三行消去一个首行元素,此时系数矩阵第三行只有一个元素
15     for j=i+1:R
16         %假设R=3, i=1时, j=2.此时内循环两次, 系数矩阵的第二三行分别消去一个首行非
17         %零元素
18         ab(j,i:C)=ab(j,i:C)-ab(i,i:C)*ab(j,i)/ab(i,i);%每一行消去首行元素
```

```

16     end
17 end
18 AB=ab;
19 for i=1:R
20     ab(i,:)=ab(i,:)/ab(i,i);
21 end
22 A
23 b
24 fprintf("高斯消去法后的增广矩阵: ")
25 AB
26 fprintf("追赶法后的增广矩阵: ")
27 ab
28
29 X=zeros(R,1);%解空间
30 X(R,1)=ab(R,C);
31 for i=R-1:-1:1
32     x(i,1)=ab(i,C)-x(i+1,1)*ab(i,i+1);
33 end
34 fprintf("解为:")
35 X
36

```

```

1  A = 4×4
2      1      2      0      0
3      2      3     -1      0
4      0      4      2     -1
5      0      0      2     -4
6
7  b = 4×1
8      7
9      9
10     10
11     12
12
13 高斯消去法后的增广矩阵:
14 AB = 4×5
15      1      2      0      0      7
16      0     -1     -1      0     -5
17      0      0     -2     -1    -10
18      0      0      0     -5      2
19
20 追赶法后的增广矩阵:
21 ab = 4×5
22     1.0000     2.0000         0         0     7.0000
23         0     1.0000     1.0000         0     5.0000
24         0         0     1.0000     0.5000     5.0000
25         0         0         0     1.0000    -0.4000
26
27 解为:
28 X = 4×1
29     7.4000
30    -0.2000
31     5.2000
32    -0.4000
33

```

3. Find the $\|X\|_1$, $\|X\|_2$, $\|X\|_\infty$

norms the following vectors

(1) $X = (3, -4, 0, \frac{3}{2})^T$,

```
1  clc;
2  clear all;
3  close all;
4  X = [3, -4, 0, 3/2]';
5  Y = [2, 1, -3, 4]';
6  [a, b, c] = Get_normal_form(X);
7  fprintf("求得的第一范式为: ")
8  a
9  fprintf("求得的第二范式为: ")
10 b
11 fprintf("求得的无限范式为: ")
12 c
13
14 function [abs_1, abs_2, abs_inf] = Get_normal_form(x)
15     abs_1 = 0;
16     abs_2 = 0;
17     for i=1:length(x)
18         abs_1 = abs_1 + abs(x(i));
19         abs_2 = abs_2 + x(i)^2;
20     end
21     abs_2 = sqrt(abs_2);
22     abs_inf = max(abs(x));
23
24 end
```

```
1  X
2  求得的第一范式为:
3  a = 8.5000
4  求得的第二范式为:
5  b = 5.2202
6  求得的无限范式为:
7  c = 4
```

(2) $Y = (2, 1, -4)^T$

```
1  Y
2  求得的第一范式为:
3  a = 10
4  求得的第二范式为:
5  b = 5.4772
6  求得的无限范式为:
7  c = 4
```